



Exploring County- Level Socioeconomic Factors Behind Voter Registration in the U.S. Rust Belt

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Background

- COVID-19 Pandemic
 - Global public-health crisis marked by widespread illness, lockdowns, and social distancing measures.
 - Unprecedented economic turmoil in supply-chain disruptions and job losses.
- The 2020 U.S. Presidential Election
 - Highly polarized and consequential for pandemic recovery, resulting in record levels of civic engagement and participation.
 - Voter registration became a more “mainstream” issue across the nation.
- Rust Belt
 - An economically diverse and politically competitive region.
 - Significant mix of deindustrialized counties with differing ranges of economic recovery.



Introduction

Research Question

- To what extent do county-level socioeconomic characteristics predict estimated voter-registration rates across Rust Belt counties in 2020?

Population

- Counties within the core states of the U.S. “Rust Belt,” focusing on:
 - Indiana, Illinois, Michigan, Ohio, Pennsylvania, and Wisconsin.

Research Objective

- Use linear regression models to identify which county-level socioeconomic factors are most strongly associated with voter-registration rates in the U.S. Rust Belt.

Data

Data for our project is collected fully from the U.S. Census Bureau.

Our dependent variable required the construction of a proxy variable that measures county-level voter registration rates.

- Constructed using CVAP 2019-2023 and Voting and Registration datasets.
- Measures county-level estimated voter registration rates among the citizen voting-age population in 2020.

Our independent variables are gathered from a collection of datasets.

- Educational attainment (S1501)
- Poverty (S1701)
- Income (S1901)
- Employment status (S2301)
- Demographic and housing estimates (DP05)

Variables



The response variable is voter registration, measured as the estimated county-level registration rate among the citizen voting-age population (i.e., CVAP) in 2020.



We will be using several explanatory variables in this experiment:

Education composition

Educational attainment levels

Economic conditions

Poverty rate

Unemployment rate

Median household income

Demographic characteristics

Age structure

Sex composition

Racial and ethnic composition

Hypothesis

- The null hypothesis indicates that there is no variability in voter registration across socioeconomic characteristics.
- The alternative hypothesis indicates that there will be significant variability in voter registration across socioeconomic characteristics.
- Our main hypothesis is that education and income will have the greatest significance on voter registration.

OLS Models

Model 0

$$reg_rate_all_pct_i = \beta_0 + \sum_{\{k=1\}}^{\{23\}} \beta_k X_{\{ki\}} + \varepsilon_i$$

Education composition

X_1 :edu_hs_plus_pct
 X_2 :edu_somcoll_pct
 X_3 :edu_ba_plus_pct
 X_4 :edu_grad_plus_pct
 X_5 :edu_lt_hs_pct
 X_6 :Percent_Less_Than_9th
 X_7 :Percent_9_12_NoDiploma
 X_8 :Percent_HS_Diploma
 X_9 :Percent_College_NoDegree
 X_{10} :Percent_Associates
 X_{11} :Percent_Bachelors
 X_{12} :Percent_Graduate

Economic controls

X_{13} :Percent_Poverty
 X_{14} :Unemployment_Rate
 X_{15} :Median_Income

Demographics

X_{16} :age_median
 X_{17} :pct_under18
 X_{18} :pct_65plus
 X_{19} :pct_male
 X_{20} :pct_female
 X_{21} :pct_hispanic
 X_{22} :pct_white_nh
 X_{23} :pct_black_nh

Model 1

$$reg_rate_all_pct_i = \beta_0 + \sum_{\{k=1\}}^{\{6\}} \beta_k X_{\{ki\}} + \varepsilon_i$$

Variables:

X1: edu_lt_hs_pct
X2: edu_ba_plus_pct
X3: Unemployment_Rate
X4: pct_under18
X5: pct_65plus
X6: pct_hispanic

Model 2

$$\log(reg_rate_all_pct_i) = \beta_0 + \sum_{\{k=1\}}^{\{9\}} \beta_k X_{\{ki\}} + \varepsilon_i$$

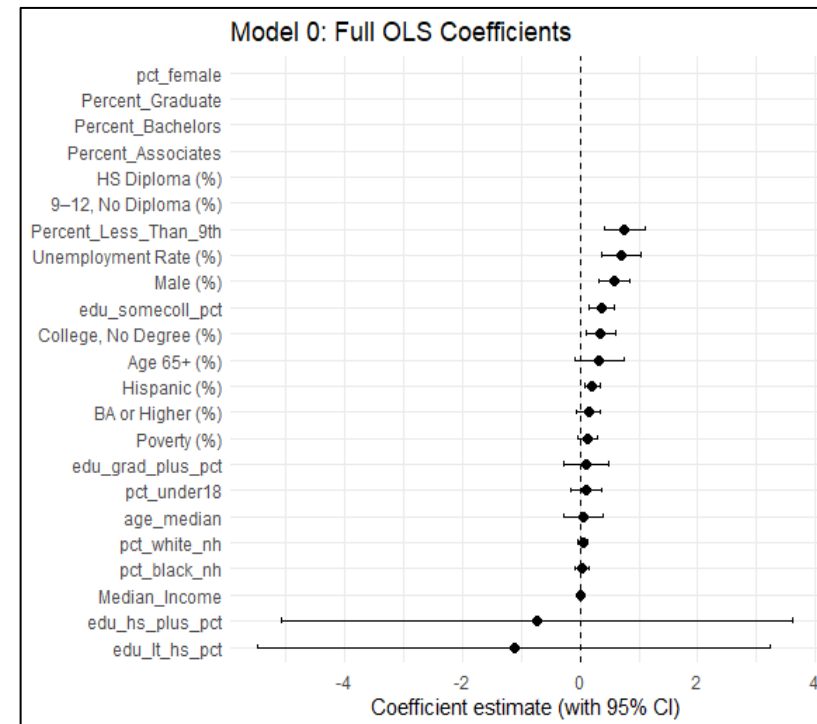
Variables:

X_1 :Percent_College_NoDegree
 X_2 :Percent_HS_Diploma
 X_3 :pct_65plus
 X_4 :Unemployment_Rate
 X_5 :pct_male
 X_6 :Percent_9_12_NoDiploma
 X_7 :edu_ba_plus_pct
 X_8 :pct_hispanic
 X_9 :Percent_Poverty

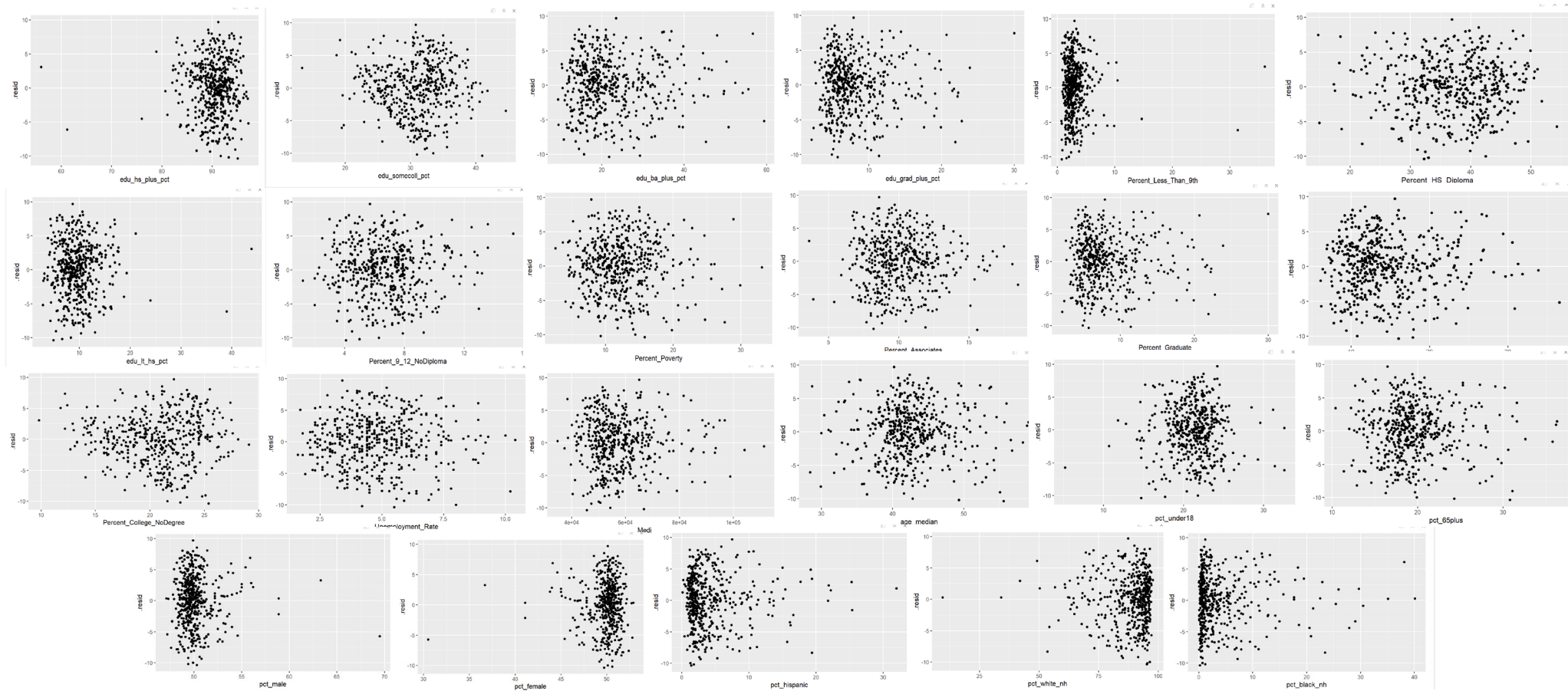
Model 0

Term	Estimate	Std Error	t value	p value	Signif
Constant	66.327	221.071	0.30	0.764	
edu_hs_plus_pct	-0.726	2.209	-0.33	0.743	
edu_somecoll_pct	0.369	0.106	3.50	0.001	***
edu_ba_plus_pct	0.141	0.100	1.41	0.160	
edu_grad_plus_pct	0.096	0.193	0.50	0.618	
edu_lt_hs_pct	-1.125	2.213	-0.51	0.611	
Percent_Less_Than_9th	0.753	0.179	4.21	0.000	***
Percent_9_12_NoDiploma	NA	NA	NA	NA	
Percent_HS_Diploma	NA	NA	NA	NA	
Percent_College_NoDegree	0.353	0.131	2.70	0.007	**
Percent_Associates	NA	NA	NA	NA	
Percent_Bachelors	NA	NA	NA	NA	
Percent_Graduate	NA	NA	NA	NA	
Percent_Poverty	0.133	0.087	1.54	0.125	
Unemployment_Rate	0.699	0.164	4.26	0.000	***
Median_Income	0.000	0.000	0.13	0.895	
age_median	0.050	0.169	0.30	0.767	
pct_under18	0.094	0.132	0.71	0.478	
pct_65plus	0.329	0.212	1.56	0.120	
pct_male	0.587	0.133	4.41	0.000	***
pct_female	NA	NA	NA	NA	
pct_hispanic	0.210	0.071	2.94	0.003	**
pct_white_nh	0.047	0.046	1.02	0.308	
pct_black_nh	0.027	0.061	0.44	0.659	

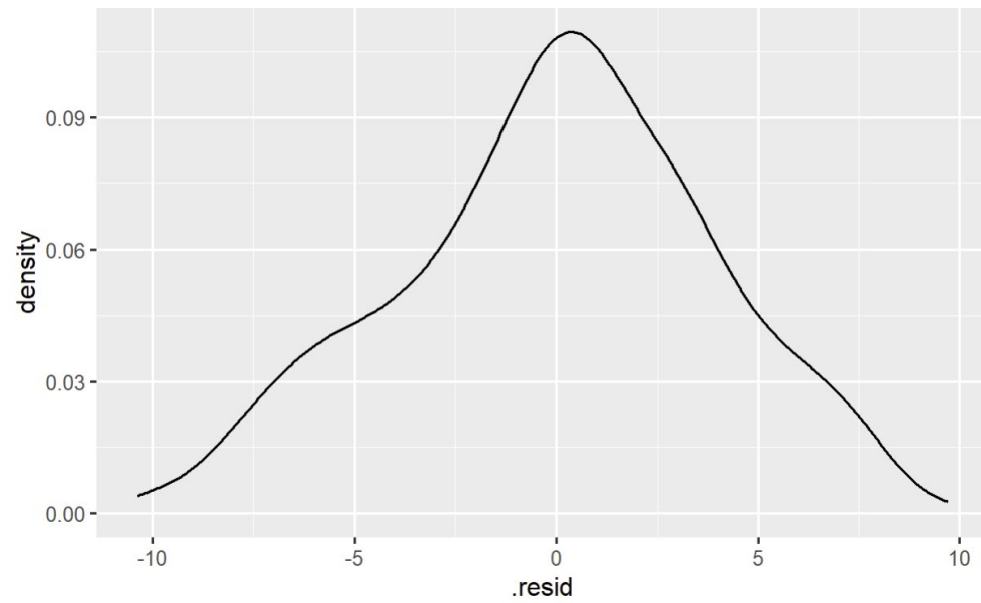
Stat	Value
N	504.000
R	0.686
R2	0.471
Adj_R2	0.452
Residual_SE	3.975
F_statistic	25.411
Model_p	0.000



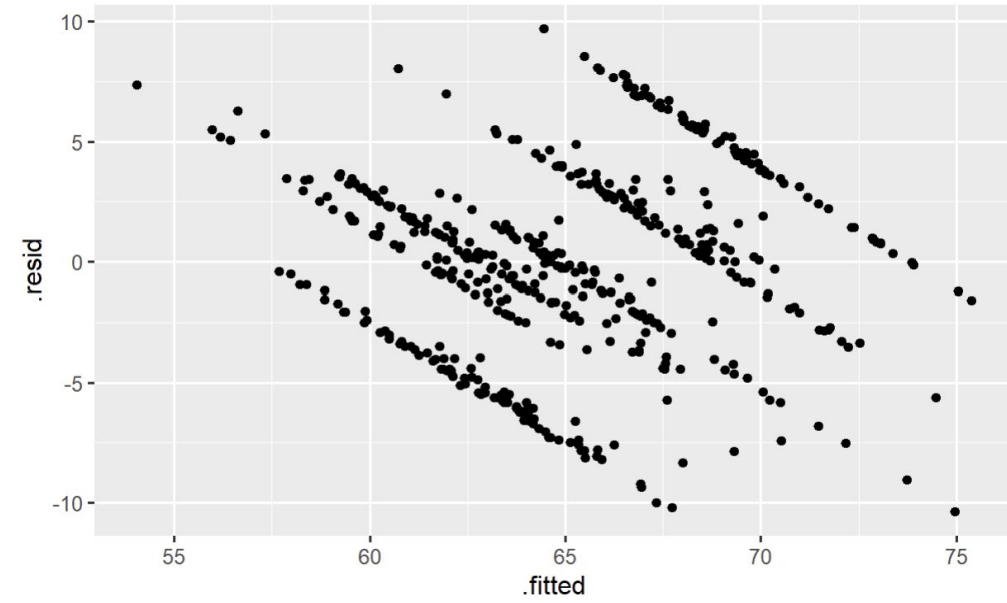
Model 0 Conditions – Linearity



Nearly Normal Residuals

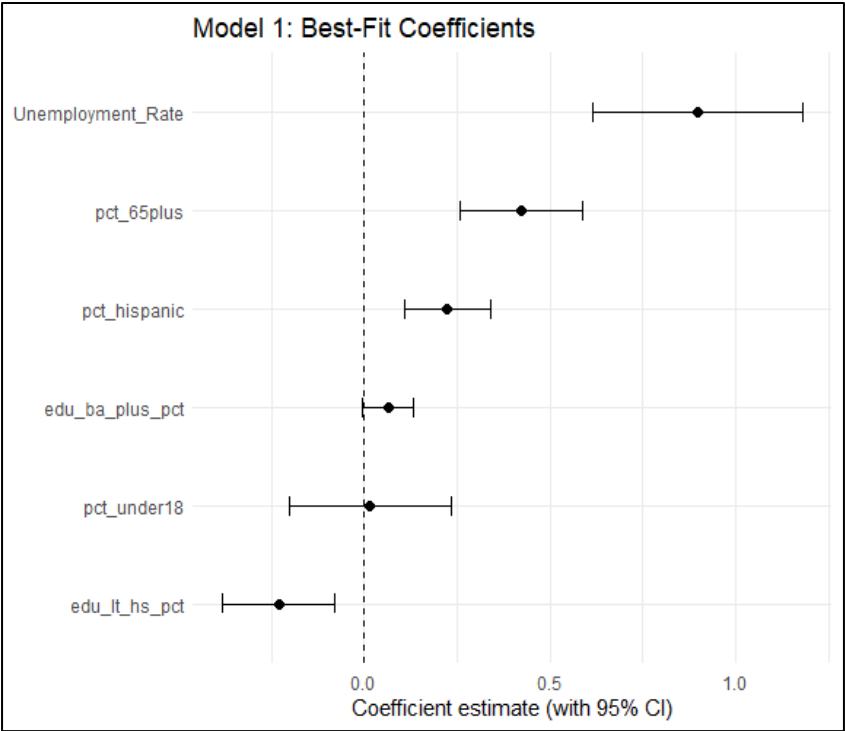


Equal Variability

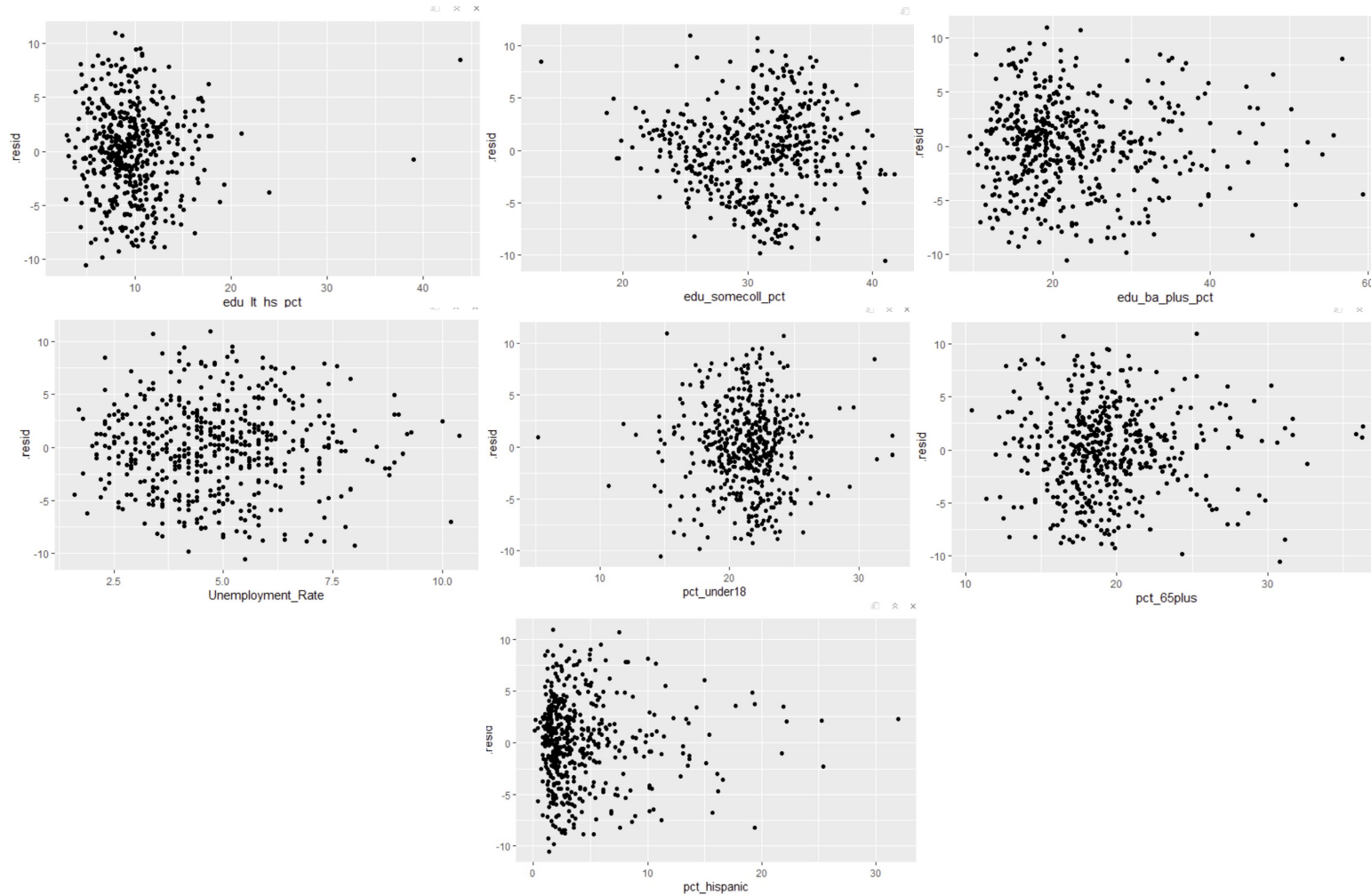


Model 1

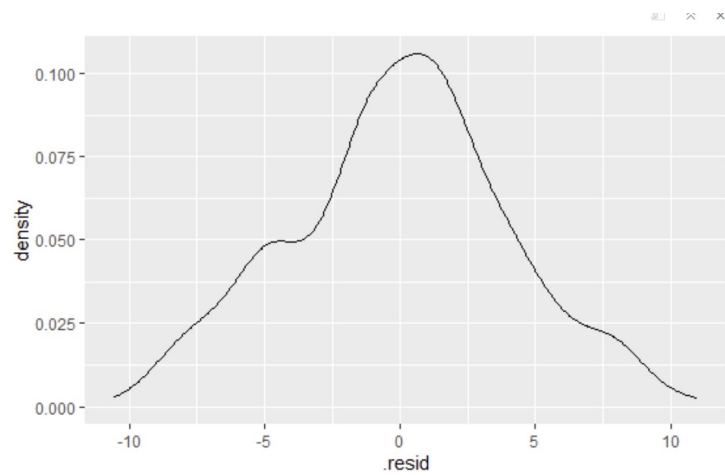
Term	Estimate	Std Error	t value	p_value	Signif	Stat	Value
Constant	52.131	4.158	12.54	0.000	***	N	504.000
edu_lt_hs_pct	-0.231	0.077	-3.01	0.003	**	R	0.459
edu_ba_plus_pct	0.064	0.036	1.81	0.071	.	R2	0.211
Unemployment_Rate	0.898	0.144	6.22	0.000	***	Adj_R2	0.201
pct_under18	0.016	0.111	0.15	0.883		Residual_SE	4.798
pct_65plus	0.424	0.084	5.05	0.000	***	F_statistic	22.151
pct_hispanic	0.224	0.059	3.77	0.000	***	Model_p	0.000



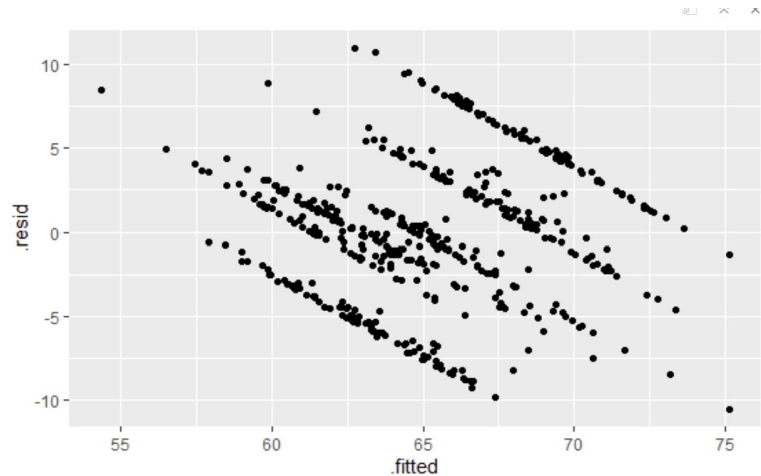
Model 1 Conditions - Linearity



Nearly Normal Residuals



Equal Variability



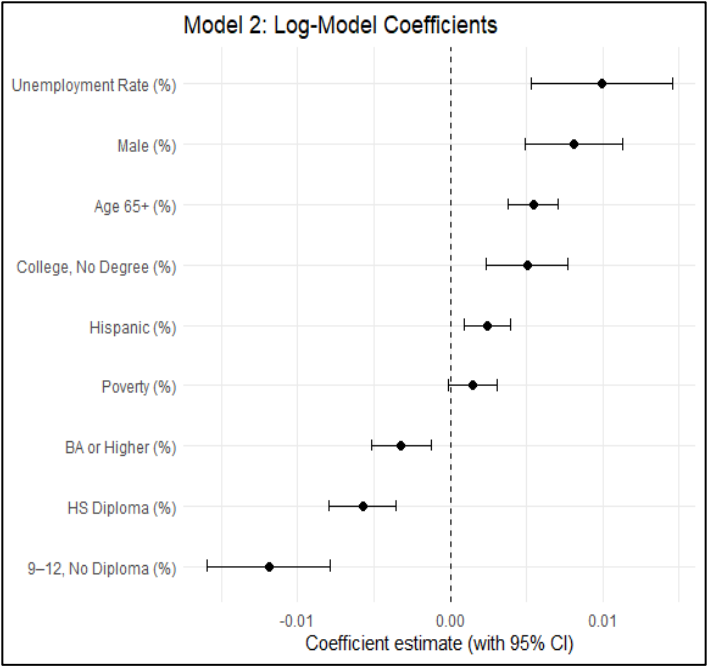
Multicollinearity

Variable	Vif() Value
edu_lt_hs_pct	2.292781
edu_somecoll_pct	1.417379
edu_ba_plus_pct	2.382770
pct_under18	2.086778
Unemployment_Rate	1.173106
pct_65plus	2.243344
pct_hispanic	1.290848

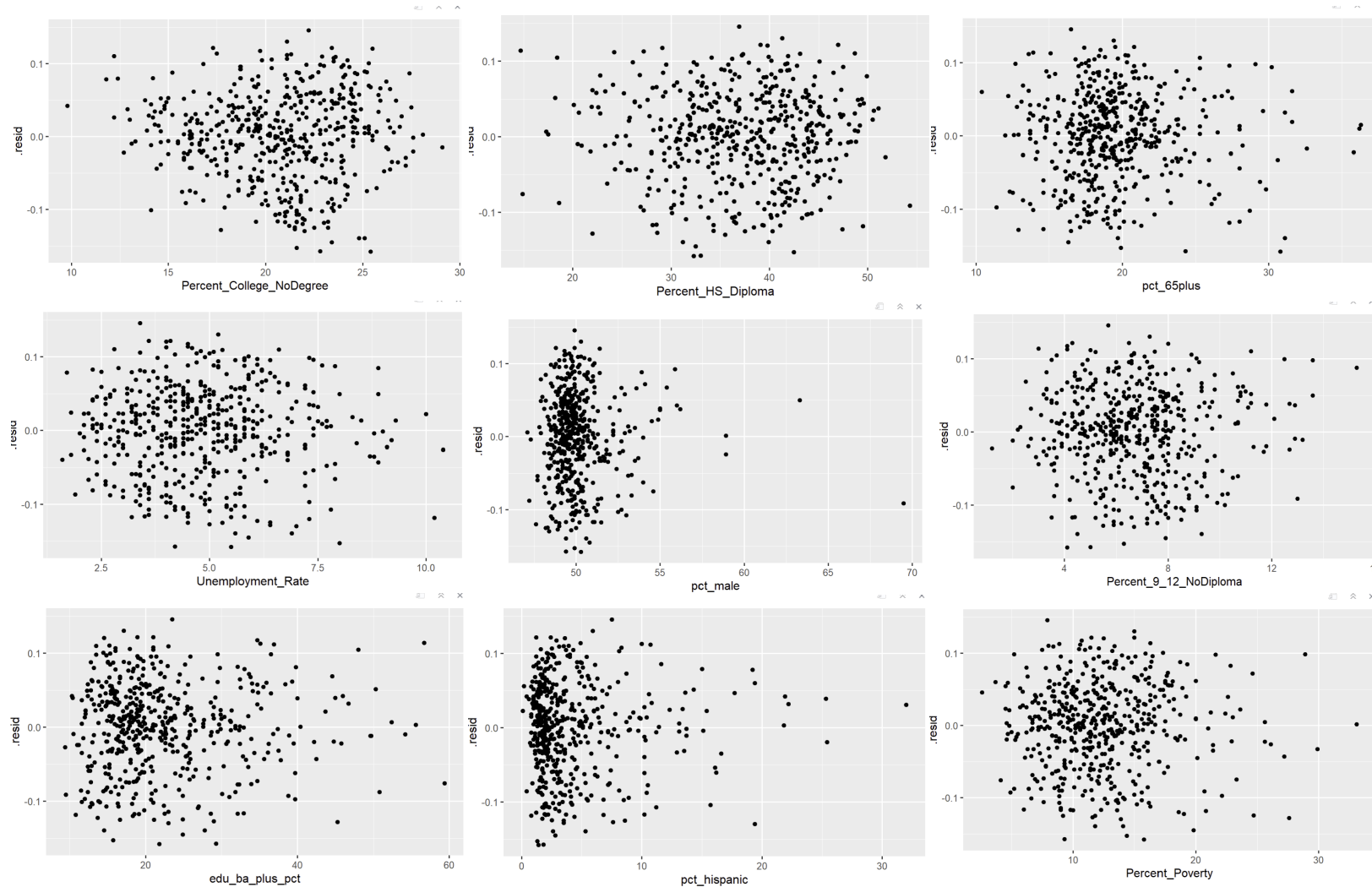
Model 2

Term	Estimate	Std Error	t value	p value	Signif
Constant	3.848	0.114	33.62	0.000	***
Percent_College_NoDegree	0.005	0.001	3.66	0.000	***
Percent_HS_Diploma	-0.006	0.001	-5.18	0.000	***
pct_65plus	0.005	0.001	6.45	0.000	***
Unemployment_Rate	0.010	0.002	4.23	0.000	***
pct_male	0.008	0.002	4.96	0.000	***
Percent_9_12_NoDiploma	-0.012	0.002	-5.82	0.000	***
edu_ba_plus_pct	-0.003	0.001	-3.25	0.001	**
pct_hispanic	0.002	0.001	3.17	0.002	**
Percent_Poverty	0.001	0.001	1.78	0.075	.

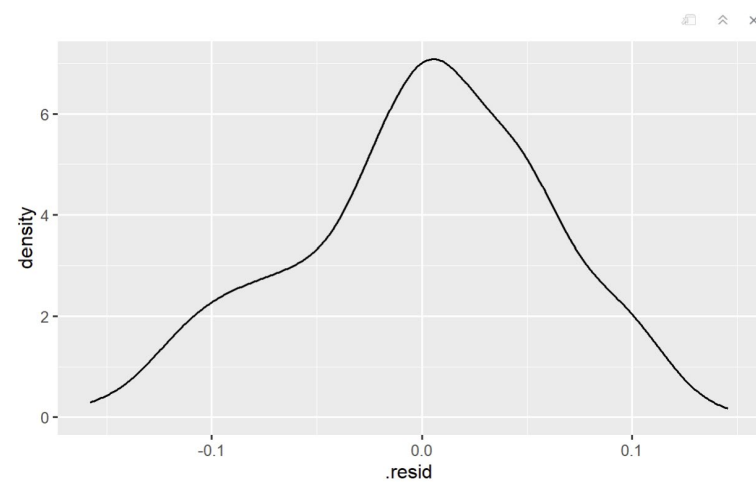
Stat	Value
N	504.000
R	0.681
R2	0.464
Adj_R2	0.454
Residual_SE	0.061
F_statistic	47.518
Model_p	0.000



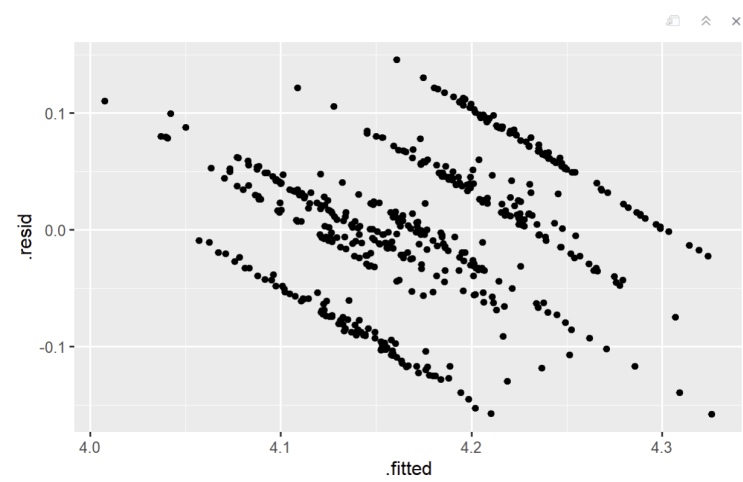
Model 2 Conditions - Linearity



Nearly Normal Residuals



Equal Variability

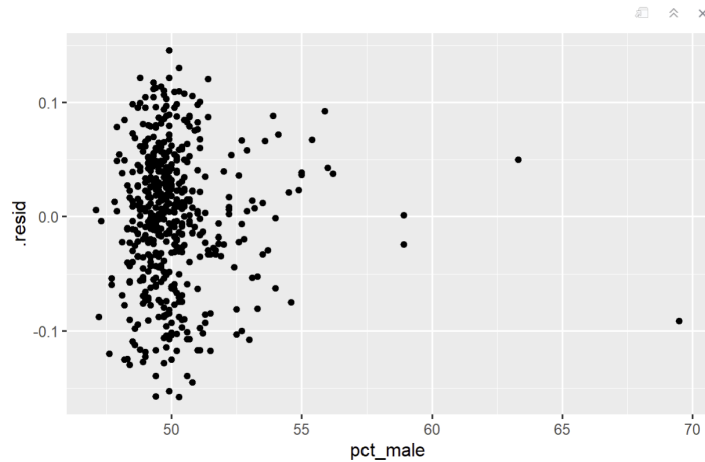


Multicollinearity

Variable	Vif() Value
Percent_9_12_NoDiploma	2.617142
Percent_College_NoDegree	2.905298
edu_ba_plus_pct	9.854644
Percent_HS_Diploma	8.139457
Percent_Poverty	1.822074
Unemployment_Rate	1.904686
pct_65plus	1.405547
pct_male	1.167809
pct_hispanic	1.332284

Limitations

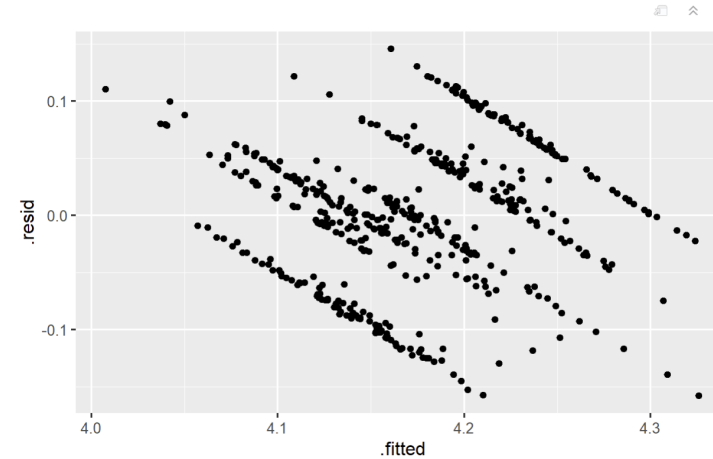
Linearity of pct_male + Residuals



Notable Outliers

- Brown County, Illinois
 - 63.3% male
- Forest County, Pennsylvania
 - 69.5% male

Equal Variance of fitted + residuals



Why are there stripes?

- Registration rates are reported at fixed decimal precision.
- Residuals therefore fall on discrete levels.
→ horizontal bands.

Is this a problem?

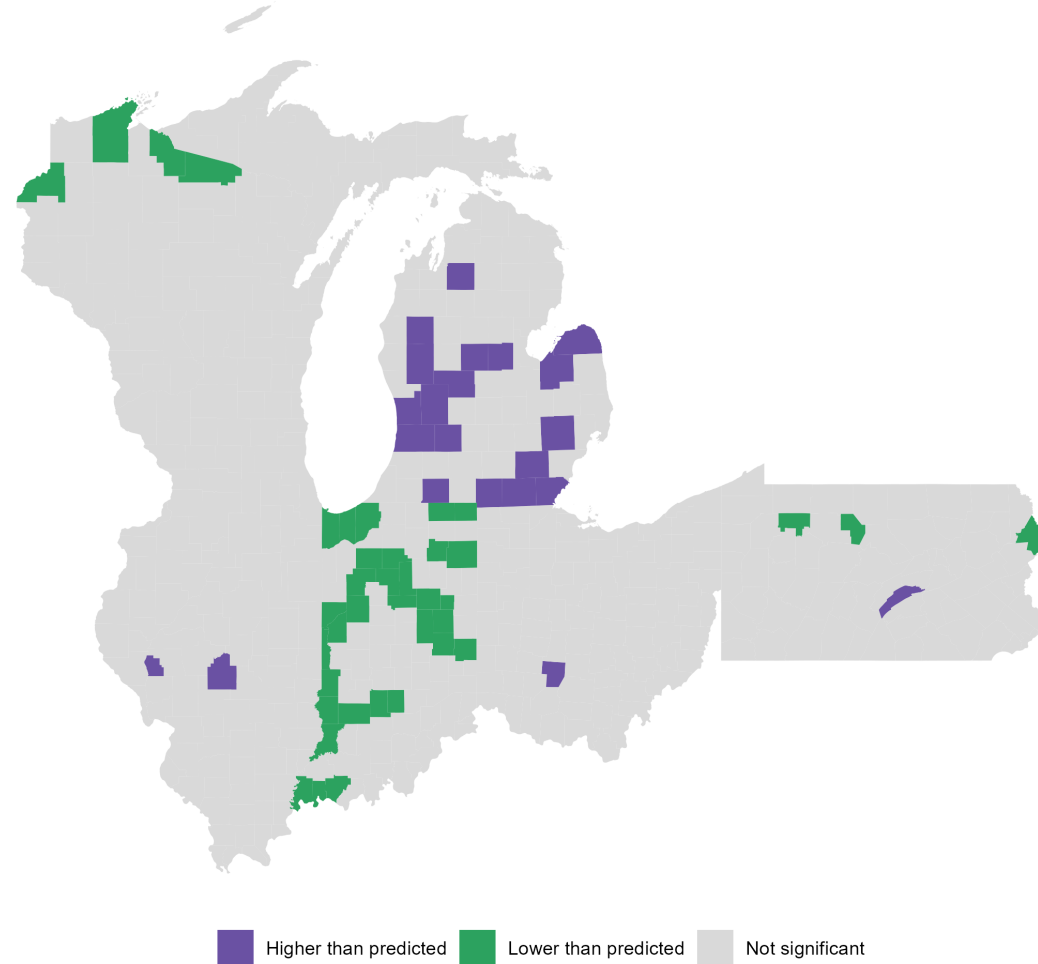
- No, the vertical spread remains constant.
- Banding is just a visual side-effect of how the data are rounded, not evidence of a modeling issue.

	Model 0: Full	Model 1: Reduced Model	Model 2: Final Model
College, No Degree (%)	0.353** SE = 0.131 p = 0.007		0.005*** SE = 0.001 p = <0.001
HS Diploma (%)			-0.006*** SE = 0.001 p = <0.001
Age 65+ (%)	0.329 SE = 0.212 p = 0.120	0.424*** SE = 0.084 p = <0.001	0.005*** SE = 0.001 p = <0.001
Unemployment Rate (%)	0.699*** SE = 0.164 p = <0.001	0.898*** SE = 0.144 p = <0.001	0.010*** SE = 0.002 p = <0.001
Male (%)	0.587*** SE = 0.133 p = <0.001		0.008*** SE = 0.002 p = <0.001
9–12, No Diploma (%)			-0.012*** SE = 0.002 p = <0.001
BA or Higher (%)	0.141 SE = 0.100 p = 0.160	0.064 SE = 0.036 p = 0.071	-0.003** SE = 0.001 p = 0.001
Hispanic (%)	0.210** SE = 0.071 p = 0.003	0.224*** SE = 0.059 p = <0.001	0.002** SE = 0.001 p = 0.002
Poverty (%)	0.133 SE = 0.087 p = 0.125		0.001 SE = 0.001 p = 0.075
Constant	66.327 SE = 221.071 p = 0.764	52.131*** SE = 4.158 p = <0.001	3.848*** SE = 0.114 p = <0.001
N	504	504	504
R2	0.471	0.211	0.464
Adj R2	0.452	0.201	0.454

Model Comparison

Model 2: Geographic Visualization

Model 2: Counties flagged by standardized residuals (90% level)



Observations and Future Research

Model Design

- Model 0 serves as our baseline to see if we recognize patterns that emerge from cleaned datasets. We use this prior to worrying about model conditions.
- Model 1 serves as our reduced model, structured as a reduced linear model to strengthen relationships.
- Model 2 serves as our final log-transformed model, placing less stress on multicollinearity while capturing non-linear relationships.

Reflection

- There is room to refine variable selection and transformations.
- One could explore alternative functional forms and different significance thresholds.

Conclusion

- Socioeconomic factors meaningfully predict voter-registration rates across Rust Belt counties.
- Both education variables (Percent_Less_than_9th) and economic variables (unemployment rate) are the strongest predictors.
- Demographic factors (Age 65+, % Male, % Hispanic) show significant associations.
- Model diagnostics indicate the results are reliable despite visually striped residuals.
- Overall, educational attainment, social demographics, and economic conditions shape county-level voting participation.

References

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